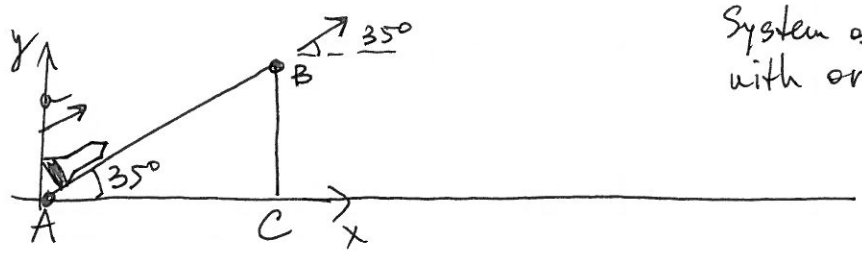
Final 2110 Solutions 12/17/09

P1



System of coordinates with origin at A.

$AC = 200\cos 35°$

$BC = 200 \sin 35°$

(a) $a = 1.25\ m/s^2 \Rightarrow V_B^2 = \overset{=0}{V_A^2} + 2a(AB) = 2(1.25)(200) = 500\ m^2/s^2$

$V_B = 22.36\ m/s$

(b) $V_x = V_{Bx} = V_B \cos 35° = 18.32$

$V_y = V_{By} - gt = V_B \sin 35° - gt$, $V_{By} = V_B \sin 35° = 12.83\ m/s$

Maximum height is reached when $V_y = 0 \Rightarrow t = \frac{V_B \sin 35°}{g}$

$$y_{max} - y_B = V_B \sin 35° \, t - \tfrac{1}{2} g t^2 =$$

$$= \frac{(V_B \sin 35°)^2}{g} - \frac{1}{2} g \left(\frac{V_B \sin 35°}{g}\right)^2 = \frac{V_B^2 \sin^2(35°)}{2g}$$

$$\Rightarrow y_{max} = 200 \sin 35° + \frac{V_B^2 \sin^2(35°)}{2g} = 123\ m$$

(c) To find horizontal range we need time of flight: this we find by solving for $y(t) = 0$

$$0 = y(t) = y_B + V_{By} t - \tfrac{1}{2} g t^2$$

$$\Rightarrow \quad t = \frac{-V_{By} \pm \sqrt{V_{By}^2 + 4(\frac{1}{2}g) y_B}}{-g} = \frac{12.83 \pm 49.12}{9.80} = 6.32\ s$$

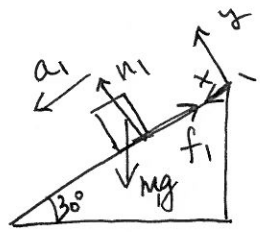
choose positive root!

Then $x_{max} = 200\cos(35°) + V_{Bx} \cdot t =$

$= 163.8 + 115.8 = 279.6\ m = 280\ m$

P2

Part 1



$$n_1 - m_1 g \cos 30° = 0$$

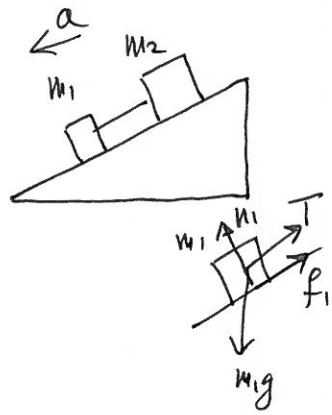
$$m_1 g \sin 30° - f_1 = m_1 a_1$$

$$f_1 = \mu_1 n_1 = \mu_1 m_1 g \cos 30°$$

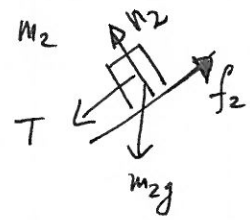
$$\Rightarrow m_1 a_1 = m_1 g \sin 30° - \mu_1 m_1 g \cos 30°$$

$$a_1 = g(\sin 30° - \mu_1 \cos 30°) = 2.78\ m/s^2$$

Part 2



m_1 and m_2 move with the same acceleration a



y-axis: $n_1 = m_1 g \cos 30°$; $n_2 = m_2 g \cos 30°$

x-axis: $m_1 g \sin 30° - f_1 - T = m_1 a$; $m_2 g \sin 30° - f_2 + T = m_2 a$

$$\Rightarrow T = m_1 g \sin 30° - \mu_1 m_1 g \cos 30° - m_1 a$$

$$\Rightarrow m_2 g \sin 30° - \mu_2 m_2 g \cos 30° + (m_1 g \sin 30° - \mu_1 m_1 g \cos 30° - m_1 a) = m_2 a$$

$$(m_1 + m_2) g \sin 30° - g \cos 30° (\mu_1 m_1 + \mu_2 m_2) = (m_1 + m_2) a$$

(a) $$a = g\left(\sin 30° - \frac{\mu_1 m_1 + \mu_2 m_2}{m_1 + m_2} \cos 30°\right) = 2.21\ m/s^2$$

(b) $T = 2.27\ N$

P3

$m = 6.00\ kg$

$x(t) = \alpha t^2 + \beta t^3$, $\alpha = 0.200\ m/s^2$, $\beta = 0.0200\ m/s^3$

(a) $v(t) = \frac{dx}{dt} = 2\alpha t + 3\beta t^2 = 2.56\ m/s$

(b) $a(t) = \frac{dv(t)}{dt} = 2\alpha + 6\beta t = 0.88\ m/s^2$

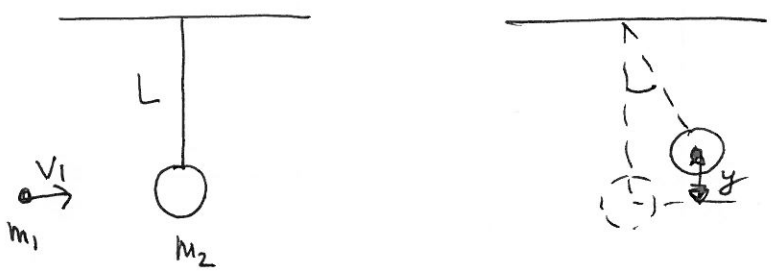
$F = ma = 5.28\ N$

(c) $W_F = K_2 - K_1$ because there is no friction

$K_1 = 0$ but $K_2 = \frac{1}{2} m v_2^2$, $v_2 = v(t = 4.00s)$

$W = \frac{1}{2} 6 (2.56)^2 - 0 = 19.7\ J$

P4



a) inelastic collision $m_1 v_1 = (m_1 + m_2) v_2$, $v_2 = \frac{m_1}{m_1 + m_2} v_1 = 0.995\ m/s$

b) energy lost $= \frac{1}{2}(m_1 + m_2) v_2^2 - \frac{1}{2} m_1 v_1^2 =$

$= \frac{1}{2}(m_1 + m_2)\left(\frac{m_1}{m_1 + m_2}\right)^2 v_1^2 - \frac{1}{2} m_1 v_1^2 =$

$= \frac{1}{2} m_1 v_1^2 \left(\frac{m_1}{m_1 + m_2} - 1\right) = -\frac{m_2}{m_1 + m_2} \frac{m_1 v_1^2}{2} =$

$= -99.5\ J$

Initial K.E. $K_1 = \frac{1}{2} m_1 v_1^2 = 100\ J$

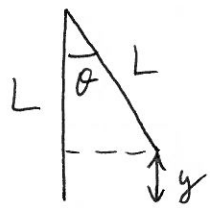
final K.E $K_2 = \frac{1}{2}(m_1 + m_2) v_2^2 = 0.5\ J$

↳ right after the collision

c) After collision energy is conserved

$$\Rightarrow \frac{1}{2}(m_1+m_2)v_2^2 = (m_1+m_2)gy$$

$$y = \frac{v_2^2}{2g} = 0.05\ m$$



$$y = L - L\cos\theta \Rightarrow \frac{y}{L} = 1-\cos\theta$$

$$\cos\theta = 1-\frac{y}{L} = 1-\frac{0.05}{0.8} = 0.937$$

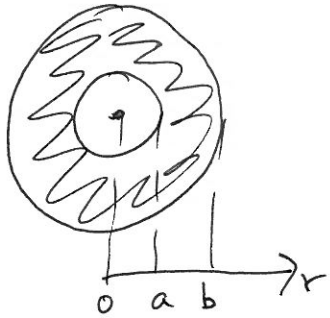
$$\theta_{max} = \cos^{-1}(0.937) = 20.4^\circ = 0.36\ rad$$

d) Angular frequency $\omega = \sqrt{\frac{g}{L}} = 3.5\ s^{-1}$

period $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}} = 1.79s = 1.8\ s$

do not depend on amplitude of oscillation
(for small enough oscillations)

P5



$\rho = 7000\ kg/m^3$

(a) Gravitational force due to planet:

$$F_g = \frac{G M_{planet} m}{b^2} = m g_{planet}$$

$$\Rightarrow g_{planet} = \frac{G M_{planet}}{b^2}$$

$$M_{planet} = \frac{4\pi}{3}\rho b^3 - \frac{4\pi}{3}\rho a^3 = \frac{4\pi}{3}\rho\left(b^3 - \left(\frac{b}{2}\right)^3\right) =$$

$$= \frac{4\pi}{3} b^3 \rho \underbrace{\left(1-\frac{1}{8}\right)}_{7/8} = \frac{7\pi}{6}\rho b^3 = \frac{7\pi}{6}(7000)(6\cdot 10^6)^3 =$$

$$= 5.539\cdot 10^{24}\ kg$$

$$g_{planet} = \frac{6.67\cdot 10^{-11}\cdot 5.54\cdot 10^{24}}{(6\cdot 10^{6})^2} = 6.026\cdot 10^{13-12} = 10.26\ m/s^2$$

pretty close to Earth's gravitational acceleration

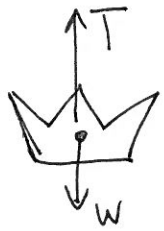
$$g_{earth} = 9.80\ m/s^2$$

(b) Inside the cavity $F_g = 0$ due to its spherical shape as we discussed in lectures.

$\boxed{P6}$

(a)

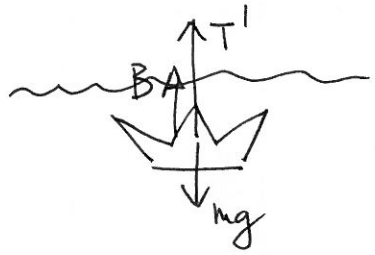
<u>in air</u>



$T - W = 0 \Rightarrow$ spring scale reads $T = W = mg$

$T = 12.9\ N$

<u>in water</u>



fully immersed

now $T + B - mg = 0$

B = weight of excluded water =

$= \rho_{water} V_{crown}\, g$

$$\Rightarrow T = -\rho_{water} V_{crown} g + mg = V_{crown}\, g\,(\rho_{crown} - \rho_{water})$$

But $m = \rho_{crown} V_{crown}$

$$T = g\,\underbrace{V_{crown}\,\rho_{crown}}_{m_{crown}}\left(1 - \frac{\rho_{water}}{\rho_{crown}}\right) = (12.9\,N)\left(1 - \frac{\rho_{water}}{\rho_{crown}}\right)$$

(b) $\rho_{crown} = \rho_{gold} \Rightarrow T = (12.9)\left(1 - \frac{1}{19.3}\right) = 12.2\ N$

(c) $\rho_{crown} = \rho_{lead} \Rightarrow T = (12.9)\left(1 - \frac{1}{11.3}\right) = 11.8\ N$